

Texture format (UVTX) - Pilotwings 64 (USA)

Findings behind the Textures tab of Hawk's Hangar. Verified on all 463 textures of the clean ROM.

File

- FORM (UVTX): two (PAD) chunks (4 bytes each) and one data chunk.
- Data chunk: (GZIP) (MIO0-compressed (COMM), 440 files) or plain (COMM) (23 files).
- Texture number = position among the UVTX files (group index), 0-462.

COMM layout (see (uvExpandTexture) in the decomp)

Offset	Type	Content
0x00	u16	image size in bytes (max 4096 = TMEM)
0x02	u16	number of display list commands
0x04	4 × f32	two texture scroll settings
0x14	...	image data = TMEM content after loading
...	8 × n	display list (G_SETTIMG, G_LOADBLOCK, G_SETTILE, G_SETTILESIZE ...)
...	u16, u16, ...	width, height, further settings

Image data

- Loaded with G_LOADBLOCK without line swapping (dxt = 0), so the data is stored in TMEM layout: rows start at multiples of 8 bytes ((line) of G_SETTILE in 64-bit words), and in every odd row the two 4-byte halves of each 8-byte word are swapped.
- Main image = tile (not 7, the load tile) with TMEM address 0, lowest tile number.
- Mipmaps = later tiles of the same format inside the image data, each smaller (315 textures, up to 5 levels).
- Some textures are combined with a second texture loaded behind the image data (second G_SETTIMG, image of another texture); those tiles lie outside the own image data.

Formats (no palette textures)

Format	Count	Notes
RGBA16	256	5-5-5 color, 1-bit alpha
I4	100	greyscale, colored by the game
I8	20	greyscale
IA4	20	3-bit grey, 1-bit alpha
IA8	58	4-bit grey, 4-bit alpha
IA16	9	8-bit grey, 8-bit alpha

Editor behavior

- Import only with the same size; the image is converted to the texture's format, mipmaps are computed from it (alpha-weighted box filter).
- An image that looks like the original (invisible pixels' color ignored) removes the project's replacement; one that looks like the current texture changes nothing.
- Recompressed MIO0 data is padded to a multiple of 4 bytes (file sizes must be multiples of 4).